ADVERTISING DECISION AUTOMATION FOR PROFITABLE TARGETING

A.D.A.P.T.

Move ad budget to where it earns profit, not just ROAS. Explained, approved by a human, and learned from.

Team M.A.R.K.A.N. · Exathon submission

ROAS says scale. The warehouse says stop

Stock blind spots

Ads keep spending on products about to sell out.

Margin blind spots

A 3× ROAS can still lose money on a thin-margin SKU.

Slow diagnosis

CTR, CVR, order value or CPM? Teams guess for days.

No feedback loop

Nobody checks if last week's move actually paid off.

COURT SNEAKER · META

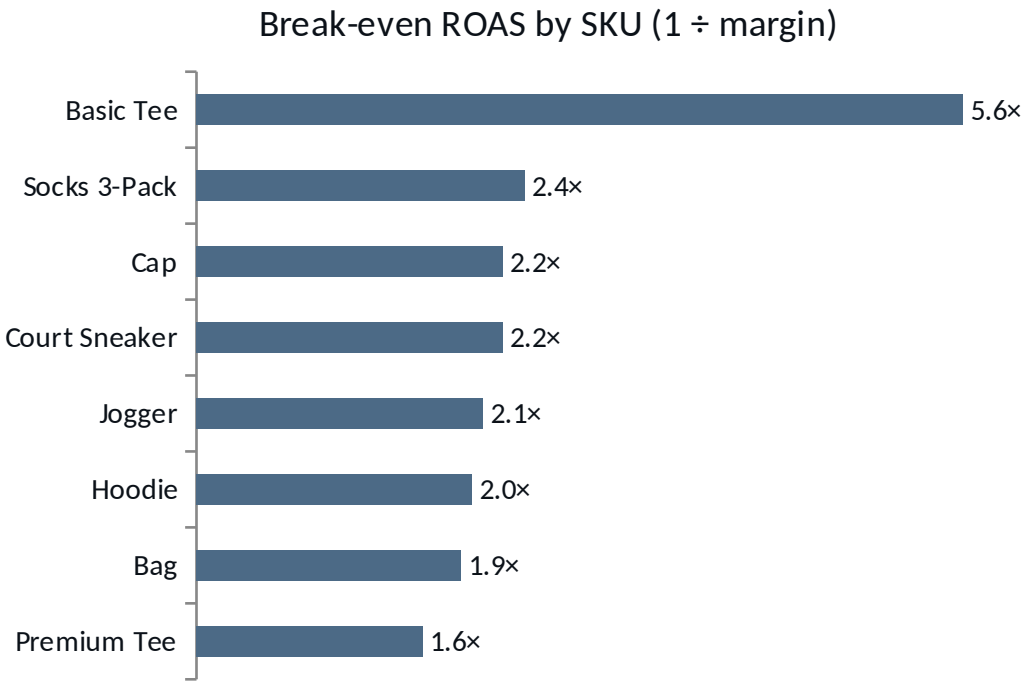
4.6×

ROAS looks great

0.9 days

of stock left (28 units)

One ROAS target fits no product



5.6×

ROAS the Basic Graphic Tee needs just to break even (18% margin)

1.6×

is enough for the Premium T-Shirt (61% margin)

Six steps from data to decision

Detect	Robust anomaly and creative-fatigue rules
Diagnose	Split ROAS change into CTR, CVR, AOV, CPM
Decide	Max profit under stock and budget guardrails
Approve	A human signs off. Every decision is logged
Simulate	3-day hidden-world run with noise and ramp
Learn	Prediction error updates model confidence

Every output is

Explained

Human-approved

Logged

Computed by code, never by an LLM

Profit math with hard guardrails

CONTRIBUTION PROFIT

```
revenue × margin − ad spend
```

BREAK-EVEN ROAS

```
1 ÷ margin_rate, per SKU
```

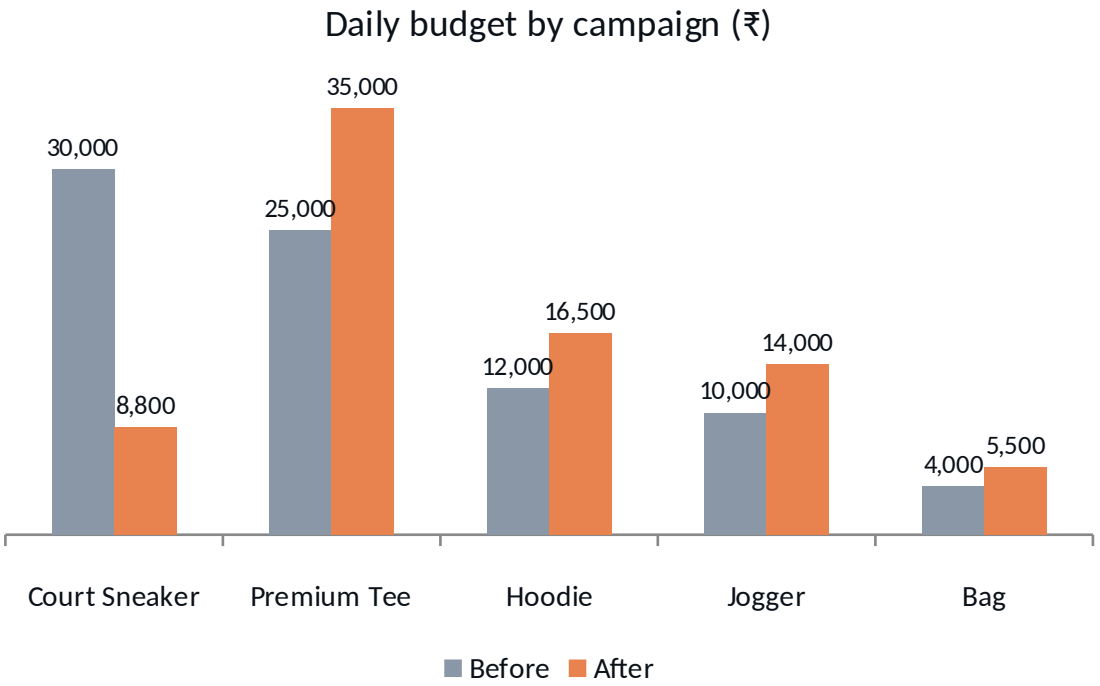
MARGINAL PROFIT PER ₹

```
margin × β × revenue ÷ spend − 1
```

Guardrails enforced in code

- Under 5 days of stock: warning. Under 3: forced donor
- Receivers keep 7 days of cover after uplift
- Moves capped at +40% / −30% per run
- Each platform keeps 60% of its spend
- Total budget conserved; leftovers held back
- Only moves worth ₹300/day or more, max 5

The plan: protect stock, fund what earns



+₹8,516

expected profit gain per day

₹1,200

held back: no profitable home

5 moves · all 7 guardrails passed · total budget conserved

The loop closes, and we show the error

₹25,549	₹74,505	191.6%	0.75 → 0.67
Predicted 3-day gain	Simulated outcome	Prediction error	Confidence after learning

Honest by design

A large miss lowers confidence by a capped step (max ±0.08). We don't tune it away.

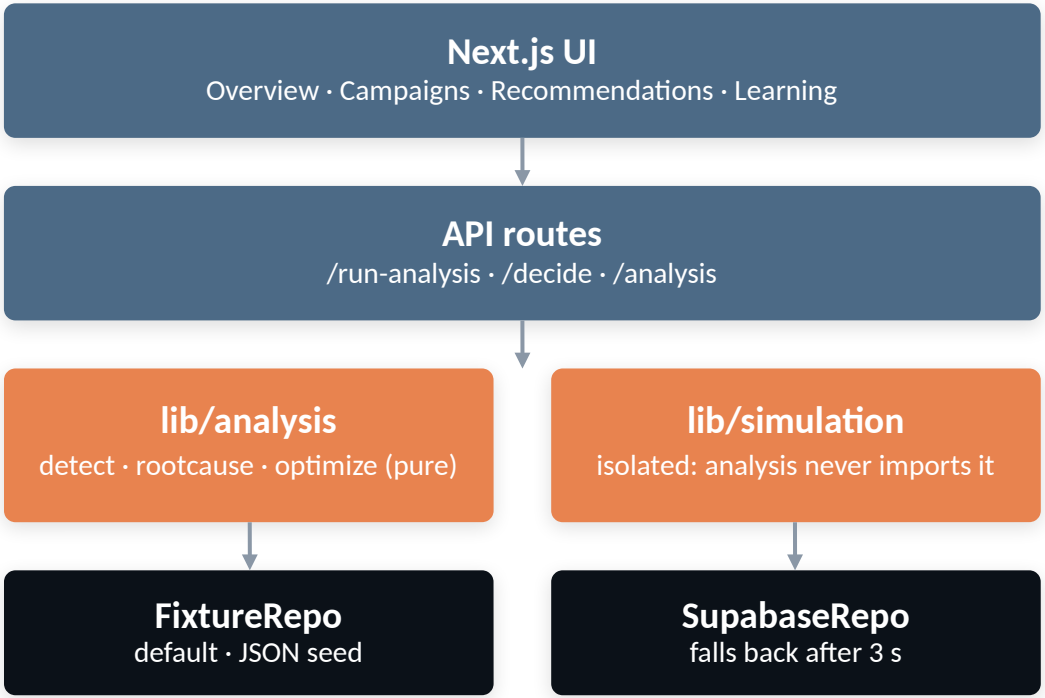
Human always decides

Approve, reject or reset. Rejected plans never execute.

Reproducible

Same seed, same result. Reset and replay for any audience.

Pure functions, one data interface



86
tests passing

~100 ms
to run a full analysis

0
LLM calls on the click path

0
real ad APIs (simulated)

From simulated to live

Connect real ad APIs

Replace simulated execution with Meta, Google and TikTok connectors.

Durable decisions

Move approval and outcome state into transactional Supabase.

More brands

Already tested on an independent skincare dataset.

github.com/Bot25692/dqps-engine · MIT License · Team M.A.R.K.A.N.